

MANDEEP KHATRI

Huntsville, AL | (202) 423-9113

mandeep.khatri@bulldogs.aamu.edu | mandyy.me | github.com/Mandeep-Khatri | linkedin.com/in/mandeep-khatri

EDUCATION

Alabama A&M University

Expected Graduation: December 2027

Bachelor of Science in Computer Science, GPA: 3.80/4.0

Huntsville, AL

- Awards & Honors:** Dean's List, AAMU Merit Full Scholarship, EICOP 2025 Finalist
- Relevant Coursework:** Data Structures & Algorithms, Intro to programming I and II (Python), Computer Architecture, Software Engineering, Operating Systems, Discrete Mathematics, Linear Algebra

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, MATLAB, Java, Go, SQL

Frameworks & Tools: React, FastAPI, Streamlit, Django, Docker, PyTest, JMeter

AI/ML & Cloud: AWS (Lambda, EC2), GCP (Vertex AI, Gemini API), PyTorch, LangChain, Kubernetes

Build/Tooling: Git, Jenkins, Playwright, Vite, Jest

Certifications: Python and Django Full Stack Web Developer Certificate, Google Cloud Platform Certificate

WORK EXPERIENCE

Academic Teaching Assistant

January 2025 – Present

Alabama A&M University

Huntsville, AL

- Assist instruction for **80+ freshmen** in **C++, Python, and Calculus I**; prepared course materials, guided labs, and graded assignments, contributing to a **15%** improvement in exam performance.
- Conduct **LeetCode-style** workshops, **live coding demos**, & project-based labs using GitHub Classroom, & **algorithm visualization tools**, mentoring students in collaborative coding and technical problem-solving.

Undergraduate Researcher – Autonomous Systems Lab

August 2024 – Present

Alabama A&M University

Huntsville, AL

- Developed and validated a **GPU accelerated vision system** for autonomous drones using **MATLAB, C++, and CUDA**, achieving a **3×** performance improvement through **optimized computation and data analysis**.
- Performed **structured testing** and **data analysis** on simulated system outputs, validating performance accuracy, **documenting results**, and **ensuring repeatable and traceable evaluation processes**.

Cloud Computing Intern

August 2023 – December 2023

Verisk

Kathmandu, Nepal

- Built a cloud-native IoT monitoring system with **Python & C++**, deploying microservices on **AWS EC2** & integrating MQTT protocol for device communication; boosted real-time data ingestion & processing efficiency by **30%**.
- Developed a responsive analytics dashboard with **HTML, CSS, & JavaScript**, backed by **Flask/Express RESTful APIs** for live sensor data; reduced user query and visualization time by **20%**.
- Conducted profiling and load testing with **PyTest, JMeter, & Dockerized** containers, resolving bottlenecks & lowering system latency by **25%**, while improving fault tolerance & aligning development with **Agile/SDLC** practices.

PROJECTS

HackMIT 2025 – Memory Tiering Simulator | C++, Python, Linux, Benchmarking

September 2025

- Developed a **two-tier memory hierarchy simulator** (fast DRAM + slow tier) during MIT's 24-hour global hackathon, implementing **hot-page detection** and **automated page migration** to reduce access latency by **30%**.
- Benchmarked performance using **YCSB workloads** and visualized **tier utilization** with **Python** to analyze **throughput** and **memory efficiency**.

LLM-Powered Course Q&A Chatbot | Python, Gemini API, LangChain, Streamlit

April 2024 – Present

- Built a Q&A chatbot using **Python, Gemini 1.5 Pro API (via Vertex AI)**, and **LangChain**, processing **100+** pages of course PDFs with **PyMuPDF**, text splitting, and embedding via **Gemini**; reached **85%** manual QA accuracy.
- Implemented **RAG** pipeline with **FAISS** (dense retrieval) and **Streamlit** UI, optimizing prompt templates and reducing avg. response latency by **40%** vs. keyword search baseline.

ACTIVITIES

Innovate Alabama AI & Public Safety Hackathon

November 2025

Google Cloud Career Jumpstart HBCU Fellow

May 2025 – August 2025

HBCU Possibilities Summit

January 2025 – May 2025

TracerFire Cybersecurity Competition - Sandia National Laboratories

September 2024